

Layer 3: Explainability

Document Owner: Zernitsky Itamar **Status:** Design Reference — Architecture Planning **Scope:** Complete specification of the explainability layer: SHAP attribution, goal attribution, counterfactual generation, confidence bands, provider risk profiles, Pareto frontier diagnostics, temporal drift detection, and cross-layer narrative generation.

What this layer does. The optimizer produces a plan. This layer makes every decision in that plan auditable, every goal outcome defensible, and every committee question answerable with a specific quantified reason. It draws simultaneously on three upstream artifacts — the XGBoost models (SHAP), the Bayesian posteriors (uncertainty bands), and the Pareto frontier (trade-off diagnostics) — to produce a complete explanation package that no single model could produce alone.

The philosophical shift. Most allocation engines produce a recommendation and stop. Layer 3 treats the recommendation as the beginning of a conversation: "here is what we recommend, here is exactly why, here is what would change the answer, here is how confident we are, and here is what to watch for next quarter." That conversation is what makes a committee trust a system enough to act on it.

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Stage 0: SHAP Attribution and Interaction Effects

What SHAP Is Doing Here

SHAP (SHapley Additive exPlanations) decomposes each XGBoost prediction into a sum of feature contributions. For provider p on day d :

$$\hat{\mu}_{\text{casetime}}(p, d) = \phi_0 + \sum_{j=1}^{27} \phi_j^{(p,d)}$$

where ϕ_0 is the global base value (mean prediction across all training samples) and $\phi_j^{(p,d)}$ is the marginal contribution of feature j to this specific prediction.

Positive ϕ_j pushes the prediction above base; negative pushes it below.

The key guarantee: SHAP values satisfy the efficiency axiom — they sum exactly to the prediction minus the base. Nothing is unaccounted for. Every minute of predicted demand is traceable to a specific feature.

SHAP DECOMPOSITION · SITE-001 · Garcia × Tuesday · Week 12			
Base value (global mean):		+187.3 min/wk	
Feature contributions (top 8, sorted by $ \phi $):			
Feature	ϕ (min)	Direction	Reason code
attn_weighted_util (0.176 vs avg 0.52)	-94.2	↓ demand	utilization_history
trailing_12w_mean_case	-12.4	↓ demand	recent_history_low
service_line_wk_mean	+2.1	↑ demand	service_line_context
trailing_4w_std_case	-1.8	↓ demand	recent_volatility
exception_rate_series	-0.9	↓ demand	series_contention
rotation_phase	+0.8	↑ demand	rotation_cycle_position
sin_woy	+0.6	↑ demand	seasonality
attn_turn_weighted	-0.1	↓ demand	turnover_history
(19 other features)	+0.6	mixed	
Prediction: 187.3 - 94.2 - 12.4 + ... = 81.6 min/wk			
HUMAN TRANSLATION:			

"Garcia's Tuesday demand is predicted at 81.6 min/wk (1.36 hrs/wk). The dominant driver pulling this below average is Garcia's chronic low utilization history (-94.2 min vs average). His recent 12-week history amplifies this further (-12.4 min). The Cardiac service line is running above average this week which slightly lifts the prediction (+2.1 min), but not enough to change the conclusion."

Reason Code Mapping

The 27 feature names are not committee-readable. Map each to a plain-language reason code:

Feature group	Features	Reason code	Plain interpretation
Attention	attn_weighted_util	utilization_history	Provider's characteristic operating level
Attention	attn_entropy	pattern_consistency	How stable the history is
Attention	attn_recency_bias	trend_direction	Whether demand is improving or declining
Attention	attn_turn_weighted	turnover_history	Characteristic turnover load
Trailing windows	trailing_4w_mean_case	recent_history	Last 4 weeks of demand
Trailing windows	trailing_12w_std_case	recent_volatility	How variable demand has been
Structural	exception_rate_series	series_contention	How contested this block slot is
Structural	service_line_wk_mean	service_line_context	What peers in the same service line are doing
Structural	rotation_phase	rotation_cycle_position	Whether this is a phase-0 or phase-1 week

Feature group	Features	Reason code	Plain interpretation
Structural	weeks_observed	data_maturity	How much history exists for this provider
Seasonality	sin_woy, cos_woy	seasonality	Time-of-year effects

SHAP Interaction Effects

The first-order SHAP values miss a critical dimension: features do not act independently. `exception_rate_series` matters more when `attn_entropy` is high — a contested slot with an inconsistent historical pattern is doubly uncertain.

SHAP interaction values Φ_{ij} decompose the prediction into main effects and pairwise interactions:

$$\hat{\mu}(p, d) = \phi_0 + \sum_j \Phi_{jj}^{(p,d)} + \sum_{i \neq j} \Phi_{ij}^{(p,d)}$$

where $\Phi_{ij} = \Phi_{ji}$ and the diagonal Φ_{jj} is the main effect after removing all interactions.

TOP SHAP INTERACTIONS · SITE-001 · CaseTime model		
Interaction pair	Mean $ \Phi_{ij} $	Interpretation
attn_weighted_util × trailing_4w_std	18.4 min	High util + high std → less predictable than either alone
exception_rate × attn_entropy	11.2 min	Contested slot with uniform history → chronic contention, not one-off event
service_line_wk_mean × rotation_phase	8.7 min	Service line context matters differently in phase 0 vs 1
weeks_observed × attn_recency_bias	6.3 min	New providers (low weeks_observed) with upward trend are more promising than trend alone suggests
ACTIONABLE INSIGHT FROM INTERACTIONS:		

Garcia has high $|\text{attn_weighted_util} \times \text{trailing_4w_std}| = 22.1$ min. His low utilization combined with high recent volatility means the model is especially uncertain about him – not just because he's low, but because his low performance is itself inconsistent. Some weeks he barely operates, others he fills half a block. This interaction is a structural signal for ZI-Gamma likelihood, not Normal.

Peer Comparison SHAP

For each provider, compute how their SHAP profile compares to their service line peers. This reveals whether a provider is an outlier or representative.

Service line aggregate SHAP: For each feature j and service line sl :

$$\bar{\phi}_j^{(sl)} = \frac{1}{|P_{sl}|} \sum_{p \in P_{sl}} \phi_j^{(p, \text{Tue})}$$

Provider deviation from peers:

$$\delta_j^{(p)} = \phi_j^{(p)} - \bar{\phi}_j^{(\text{service_line}(p))}$$

PEER COMPARISON SHAP · Cardiac service line · Tuesday slot

Feature	Garcia ϕ	Cardiac avg ϕ	δ (Garcia vs peers)
utilization_history	-94.2	+38.1	-132.3 ← massive gap
recent_history	-12.4	+9.8	-22.2
service_line_context	+2.1	+2.1	0.0 (shared)
series_contention	-0.9	+0.4	-1.3
recent_volatility	-1.8	-0.6	-1.2

DIAGNOSIS: Garcia's utilization_history SHAP is 132 min below his Cardiac peers. This is not a model artefact – it directly reflects that Garcia's attn_weighted_util (0.176) is far below the Cardiac service line mean (0.72). The model has correctly identified Garcia as structurally different from his peers, not just temporarily low.

IMPLICATION FOR THE COMMITTEE: The question to ask is not "what is Garcia's target utilization?" but "why is Garcia operating at 1/4 of his Cardiac peers' level?" Layer 3 surfaces this question. It does not answer it – that is a clinical and operational conversation.

Stage 1: Goal Attribution

Connecting the Chance Constraint Math to Block Changes

For each goal provider, explain exactly why the engine made the specific changes it did — from the $R[p,s]$ matrix through the chance constraint to the block-level actions.

Block Removal Order

Blocks are removed in ascending order of combined (CaseTime + Turnover) utilization:

$$\text{util_combined}[p][b] = \frac{\text{CaseTime}_{p,b} + \text{Turnover}_{p,b}}{\text{AllocatedBlockTime}_{p,b} - \text{EarlyRelease}_{p,b}}$$

The lowest-utilization blocks are removed first. This is not a heuristic — it is the optimal order under O1 (deviation minimization): removing the lowest-utilization blocks minimizes the loss in allocated time relative to actual demand usage.

Coverage Curve

For each feasible block count $n \in 1, 2, 3, \dots$, compute the fraction of the 200 scenarios covered:

$$\text{coverage}(n) = \frac{1}{S} \sum_{s=1}^S \mathbf{1} [n \cdot \text{block_hours} \geq R[p, s]]$$

Surface this as a table so the committee sees exactly what each additional block buys in coverage probability.

GOAL ATTRIBUTION · Dr. Garcia · Target utilization 70%				
DEMAND ANALYSIS				
CaseTime demand band:	P10=0.78h	P50=1.36h	P90=1.87h	per week
Turnover demand band:	P10=0.11h	P50=0.22h	P90=0.32h	per week
Total demand band:	P10=0.96h	P50=1.55h	P90=2.11h	per week
Quarterly total P50:	20.1 hrs			
EarlyRelease projected:	5.3 hrs	(deterministic)		

COVERAGE CURVE (target=70%, block_hours=8 hrs each)

n blocks	Allocated	Scenarios	Coverage	Meets $\alpha=0.85$?
1	8 hrs	2/200	1%	x
2	16 hrs	3/200	2%	x
3	24 hrs	24/200	12%	x
4	32 hrs	79/200	40%	x
5	40 hrs	148/200	74%	x (very close)
6	48 hrs	174/200	87%	✓ ← minimum feasible
7	56 hrs	191/200	96%	✓

NOTE: The minimum feasible allocation is 6 blocks (48 hrs) – covering 87% of scenarios. This is much higher than the DEMO-SITE example (2 blocks) because SITE-001 real Garcia has different demand characteristics.

BINDING FACTOR: target_level (increasing to 6 blocks achieves coverage – demand is sufficient, target is achievable with correct block count)

BLOCK REMOVAL ORDER (28 → 6 blocks, removing 22):

Rank	Block ID	Combined util	Decision
1	BLK-087	1.8%	REMOVE (lowest util in portfolio)
2	BLK-091	2.1%	REMOVE
3	BLK-094	2.3%	REMOVE
...	...	...	REMOVE
22	BLK-044	12.4%	REMOVE
23	BLK-038	15.7%	KEEP (sufficient for 6-block coverage)
...	...	...	KEEP
28	BLK-011	31.2%	KEEP (highest util – anchor block)

Stage 2: Counterfactual Generation and What-If Studio

Standard Counterfactuals

For each utilization-goal provider, compute how the allocation changes under alternative targets:

$$R_{\text{alt}}[p, s] = \frac{Q_{\text{casetime}}^{(s)}[p] + Q_{\text{turnover}}^{(s)}[p]}{u} + \text{EarlyRelease}_{\text{projected}}[p]$$

$$\text{coverage_alt}(u) = \frac{1}{S} \sum_{s=1}^S \mathbf{1} [A[p] \geq R_{\text{alt}}[p, s]]$$

The What-If Studio

Beyond simple target changes, the What-If Studio allows the committee to ask any structured counterfactual question without re-running the full MIP. Each question type maps to a specific computation over the existing scenario library.

Type 1 — What if the target changes?

Already covered above. $O(S)$ computation — instant.

Type 2 — What if we give Garcia one more block?

Recompute coverage at $A[p] + \text{block_hours}$:

$$\text{coverage_plus1} = \frac{1}{S} \sum_{s=1}^S \mathbf{1} [A[p] + 480 \geq R[p, s]]$$

No MIP re-solve needed. The coverage jump from block n to $n + 1$ is immediately visible.

Type 3 — What if Garcia's demand increases by 20% next quarter?

Scale all 200 scenarios by 1.20 and recompute coverage:

$$Q_{\text{scaled}}^{(s)}[p] = 1.20 \times Q_{\text{total}}^{(s)}[p]$$

$$R_{\text{scaled}}[p, s] = Q_{\text{scaled}}^{(s)}[p] / \text{TargetUtilization} + \text{EarlyRelease}$$

This is a demand sensitivity analysis. The committee can ask: "if Garcia starts ramping up his practice, at what point does our allocation become insufficient?"

The answer is the demand multiplier at which coverage drops below 85%.

Type 4 — What if Wright's range_available starts earlier?

This changes the feasible set for Wright — more blocks become eligible. This requires a partial MIP re-solve but only for Wright's variables (a much smaller problem). Run a single-provider sub-MIP in under 2 seconds.

Type 5 — What if we switch to a different Pareto theme for this provider?

Recompute the single-provider outcome under each theme without changing the rest of the allocation. This answers: "if we had run continuity_first for Garcia specifically but utilization_first for everyone else, what would Garcia's allocation look like?"

WHAT-IF STUDIO · Garcia · Selected counterfactual questions

Q1: What target achieves coverage with only 4 blocks?

A: 4 blocks (32 hrs) → coverage=40%.

Need coverage $\geq 85\%$ at 32 hrs. The target that achieves this:

$R[\text{Garcia}, s] = 32$ for 85% of scenarios

$P85 \text{ of } (Q_{\text{total}}^s) = 27.4 \text{ hrs/qtr}$

$27.4 / 0.70 + 5.3 = 44.5 \text{ hrs} \rightarrow \text{impossible at 32 hrs}$

Solve for u : $27.4 / u + 5.3 = 32 \rightarrow u = 27.4 / 26.7 = 1.026 \rightarrow >100\%$

Conclusion: no feasible target achieves 85% coverage at 4 blocks.

Garcia needs at least 6 blocks regardless of target level.

Q2: What demand increase would require a 7th block?

A: At 6 blocks (48 hrs), coverage = 87%.

Scale Q_{total}^s by multiplier λ until coverage drops below 85%:

$\lambda = 1.08 \rightarrow \text{coverage} = 84\% \rightarrow \text{the 7th block becomes necessary.}$

Interpretation: if Garcia's surgical volume grows by 8% quarter-over-quarter, his 6-block allocation will become insufficient.

Q3: What if Garcia's early release pattern changes (reduces to 0)?

A: $\text{EarlyRelease}_{\text{projected}} = 0$ instead of 5.3 hrs.

$R[\text{Garcia}, s] = Q_{\text{total}}^s / 0.70 + 0$ (no early release term)

Minimum feasible blocks: 5 (instead of 6). One block fewer needed

if Garcia stops releasing blocks early. This is actionable feedback for the OR director: "coaching Garcia on block retention could reduce his allocation need by one block."

Q4: What if Garcia's σ_{pd} were as tight as Chen's (8.3 vs 25.5 min)?

A: Regenerate scenarios with $\sigma_{\text{pd}} = 8.3$ for Garcia.

Scenario spread narrows: $P10=1.12\text{h/wk}$ $P50=1.36\text{h/wk}$ $P90=1.62\text{h/wk}$

Minimum feasible blocks drops from 6 to 5.

Interpretation: Garcia's demand uncertainty costs him 1 block.

Reducing variability (consistent scheduling, fewer ad-hoc changes) would be worth as much as a small demand increase.

Reverse Counterfactuals

Standard counterfactuals ask: "given this provider, what changes?" Reverse counterfactuals ask: "what would this provider need to look like for the answer to change?"

For each goal provider, find the minimum change to any feature that would change the recommended block count by ± 1 :

$$\min_{\delta \in \mathbb{R}} |\delta| \quad \text{s.t.} \quad \text{coverage} \left(n_{\text{current}} - 1 \mid \hat{\mu}' = f(\mathbf{x} + \delta \mathbf{e}_j) \right) \geq 0.85$$

This is computed by scanning δ over a grid for the most impactful feature (highest $|\phi_j|$). The result is: "Garcia would need his attn_weighted_util to increase from 0.176 to 0.31 — equivalent to doubling his OR activity — for the engine to recommend 5 blocks instead of 6."

These reverse counterfactuals are particularly valuable for clinical conversations: they translate the model's decision into a behavioral target for the provider.

Stage 3: Confidence Bands and Provider Risk Profiles

Confidence Bands on Projected Utilization

For each scenario s , compute projected utilization under the proposed allocation:

$$\text{util}^{(s)}[p] = \frac{Q_{\text{casetime}}^{(s)}[p] + Q_{\text{turnover}}^{(s)}[p]}{A[p]}$$

Report P10, P25, P50, P75, P90 across $S=200$ scenarios. The band width reveals more than the P50 alone.

UTILIZATION BANDS · SITE-001 · Selected providers								
Provider	Blocks	P10	P25	P50	P75	P90	Band	Risk
Garcia	6→6	42%	55%	68%	81%	97%	55pp	△ HIGH.
Chen	40→40	87%	89%	91%	93%	95%	8pp	✓ LOW
Thompson	14→5	58%	63%	69%	76%	82%	24pp	✓ MED
Wright	0→2	—	—	—	—	—	N/A	NEW
Kim	14→14	38%	42%	45%	48%	52%	14pp	△ LOW

BAND WIDTH INTERPRETATION:

Garcia band=55pp: The difference between a good quarter (P90=97%) and a bad quarter (P10=42%) is enormous. His target of 70% is achievable in the median scenario but the variance means ~40% of quarters he will miss it. The committee should understand this is a probabilistic target, not a guarantee. The band IS the honest answer.

Chen band=8pp: Nearly certain to operate at 87-95% regardless of scenario. High confidence, predictable outcome.

Kim band=14pp, P50=45%: Kim has 14 blocks but only uses 41% of them. His band is tight meaning this is consistent underuse, not variability. No goal was set on Kim – but Layer 3 surfaces this as a potential opportunity for a future goal: Kim's blocks might be reclaimed.

Provider Risk Profiles

Combine four signals into a 2×2 risk classification per provider:

Signal 1 — Demand uncertainty (from Bayesian model): σ_{pd} posterior mean. High σ = uncertain demand.

Signal 2 — Utilization band width (from Layer 3): P90 – P10 of projected utilization. Wide band = uncertain outcome.

Signal 3 — Binding factor (from Layer 2): case_volume / target_level / supply_shortage — why the goal may not be met.

Signal 4 — SHAP dominance (from Stage 0): which reason code has the largest absolute SHAP value. Structural drivers (utilization_history, series_contention) are more stable than transient ones (seasonality, recent_volatility).

PROVIDER RISK PROFILES · SITE-001 · Q3 2026

RISK MATRIX (demand uncertainty × outcome uncertainty)

	Low outcome uncertainty (band < 20pp)	High outcome uncertainty (band ≥ 20pp)
Low demand uncertainty	Chen, Kim, Jones "Predictable performers"	Torres, Patel "Variable performers"

$(\sigma_{pd} < 15 \text{ min})$	→ stable plans, reliable outcomes, safe to give aggressive targets	→ wider bands, may miss targets
High demand uncertainty $(\sigma_{pd} \geq 15 \text{ min})$	Rivera, Moore "Uncertain but stable" → demand hard to predict but outcome bands are acceptable	Garcia, Thompson, Wright "High attention needed" → both demand and outcome uncertain, goals need close monitoring
QUADRANT IMPLICATIONS:		
"Predictable performers": Safe to set aggressive targets. The engine's allocation for these providers is highly reliable – P50 and actual will likely align within 5pp.		
"Variable performers": Wide band means the goal is probabilistic. The committee should set targets knowing ~30% of quarters will miss.		
"Uncertain but stable": The demand model is uncertain (wide σ_{pd}) but the provider's behavior, once they start operating, is consistent. The uncertainty is in when/whether they operate, not in how much.		
"High attention": Garcia and Thompson. Goals are set, blocks are allocated, but both the demand and the outcome are genuinely uncertain. These are the providers to review first each quarter.		

Underuse and Capacity Flags

Two binary flags surface automatically for every provider:

$\text{underuse_risk_flag}[p] = 1$ [P50 projected util < 0.50]

$\text{capacity_risk_flag}[p] = 1$ [P90 projected util > 1.00]

underuse_risk_flag means: even in the median scenario, this provider will use less than half their allocated time. The allocation is correct given the probabilistic guarantee, but the committee should have low expectations for the utilization number.

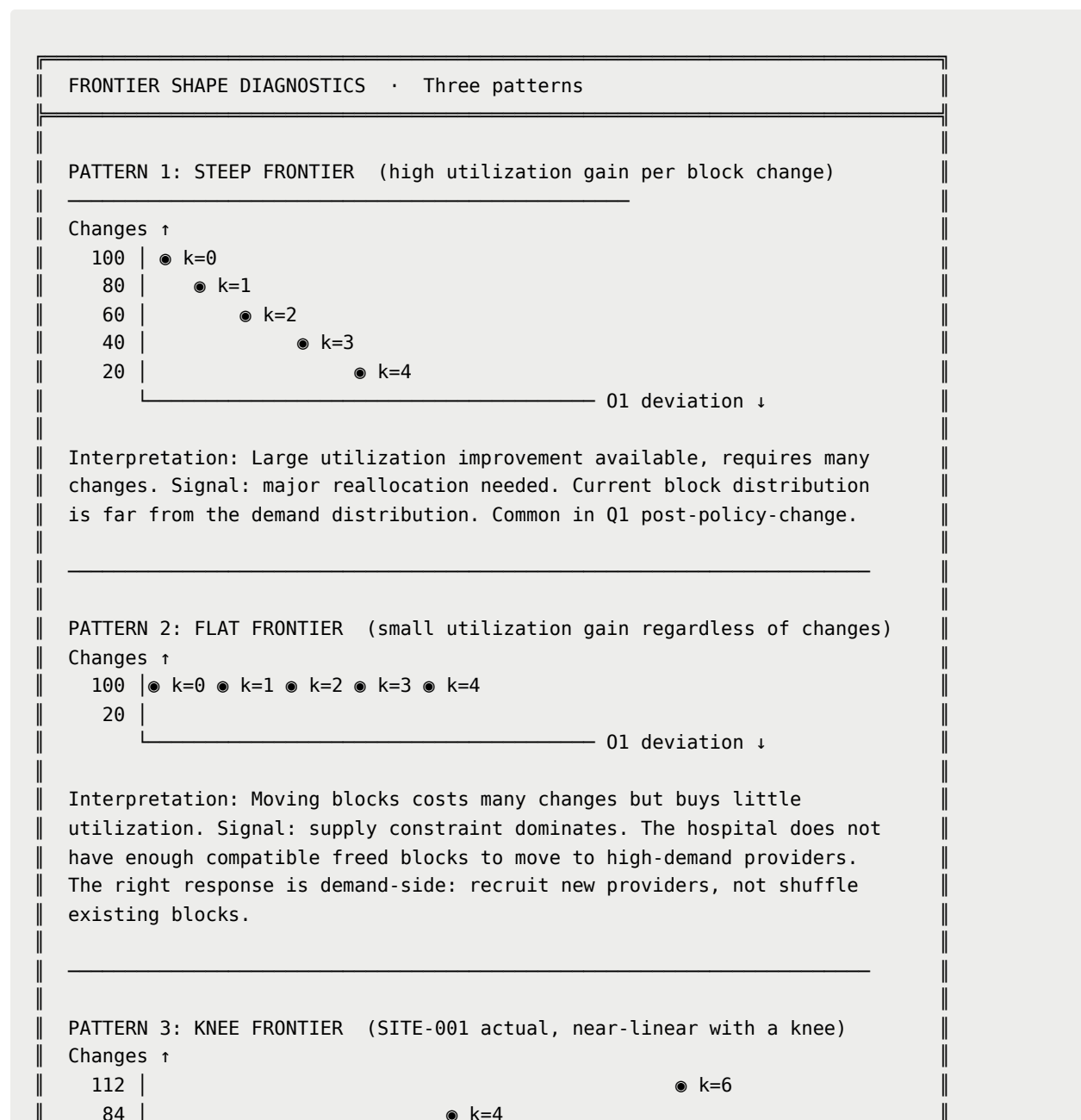
capacity_risk_flag means: in busy scenarios, this provider will overflow their blocks — demand exceeds capacity in the top 10% of futures. For elective OR scheduling this is manageable (cases can be deferred), but the flag prompts the committee to discuss whether overflow blocks are available.

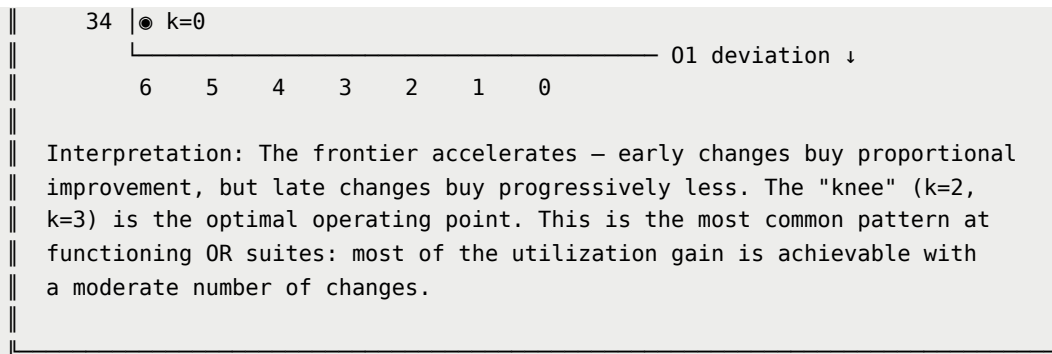
Stage 4: Pareto Frontier Diagnostics

The Frontier as a Diagnostic Instrument

The shape of the Pareto frontier in (O1, O3) space carries information about the operational structure of the hospital that no single-objective solve can reveal.

Three frontier shapes have distinct interpretations:





Frontier Shift Tracking

Track the frontier across quarters to detect operational trends:

Frontier expanding outward (Q2 frontier dominates Q1 frontier): the hospital has become more flexible — more blocks can be reallocated without disrupting continuity. Usually follows a policy change that reduced block tenure protections.

Frontier contracting inward (Q1 frontier dominates Q2 frontier): the hospital has become more constrained — OR supply is tighter, provider availability is more restricted. Early warning of capacity strain before it becomes a crisis.

Frontier rotating (Q2 is better on utilization but worse on stability): the hospital has new demand from new providers (Wright ramping up), which makes utilization improvement available but requires touching more incumbent blocks.

SHAP Drives and Frontier Structure

One of the more interesting cross-layer insights: there is a direct connection between the top SHAP drivers in Layer 1 and the shape of the Pareto frontier in Layer 2.

When utilization_history dominates SHAP (Garcia's case): the demand uncertainty is structural — the provider simply does not generate enough cases. The frontier will be steep: many blocks need to move because the mismatch between demand and allocation is large.

When series_contention dominates SHAP: multiple providers are competing for the same slots. The frontier will be flat on the utilization dimension but show rapid

improvement in the continuity dimension — the value of optimization is not utilization improvement but sorting out who owns which recurring slot.

When seasonality dominates SHAP: demand is time-varying and the optimization is time-sensitive. The frontier will be different at Q1 (winter, lower volume) vs Q3 (summer, higher volume for elective procedures). Comparing Q1 and Q3 frontiers reveals the seasonal amplitude.

Stage 5: Temporal Drift Detection

Why Drift Matters

The XGBoost model and Bayesian posterior are fit on historical data. If a provider's behavior changes — Garcia starts operating more consistently, a new service line is introduced, the OR schedule policy changes — the model's beliefs become stale. Stale beliefs produce miscalibrated scenarios. Miscalibrated scenarios produce unreliable chance constraint guarantees.

Drift detection is the mechanism that catches this before the committee gets a recommendation based on outdated data.

Three Drift Signals

Signal 1 — SHAP Temporal Drift

For each provider and each feature, track the SHAP value across quarters. A SHAP value that was stable for two years and suddenly changes by >50% signals a structural change in that feature's relationship to demand.

$$\text{SHAP_drift}[p][j] = \frac{|\phi_j^{(p,t)} - \phi_j^{(p,t-1)}|}{|\phi_j^{(p,t-1)}|}$$

Flag when $\text{SHAP_drift} > 0.5$ for any feature contributing more than 10 min to the prediction.

What it catches: A provider who starts generating more turnover (new case mix, longer procedures) will show SHAP_drift on `attn_turn_weighted`. A provider who starts using their blocks more consistently will show drift on `attn_entropy`.

Signal 2 — Posterior σ_{pd} Drift

Compare the posterior mean of σ_{pd} across quarters:

$$\text{sigma_drift}[p][d] = \frac{|\sigma_{pd}^{(t)} - \sigma_{pd}^{(t-1)}|}{\sigma_{pd}^{(t-1)}}$$

A narrowing σ_{pd} means a provider is becoming more predictable — their demand is stabilizing. A widening σ_{pd} means they are becoming more erratic — their pattern is breaking down.

What it catches: Wright's σ_{pd} should narrow over time as he accumulates history. If it does not narrow (if it stays wide even after 4 quarters), this signals that Wright's demand is genuinely volatile, not just uncertain due to sparse data. Different clinical implication.

Signal 3 — Calibration Drift

Compare the empirical coverage from the calibration check quarter-over-quarter:

$$\text{coverage_drift}[L] = \text{coverage}_L^{(t)} - \text{coverage}_L^{(t-1)}$$

A systematic decline in empirical coverage at all three nominal levels (50%, 80%, 95%) signals that the model is becoming overconfident — the scenarios are too narrow relative to actual outcomes.

DRIFT DETECTION REPORT · SITE-001 · Q3 2026 vs Q2 2026					
SHAP TEMPORAL DRIFT (providers with any feature drift > 50%)					
Provider	Feature	Q2 ϕ	Q3 ϕ	drift	Action
Wright	attn_weighted_util	0.000	+18.3	∞	← expected (first real data – prior shifts to posterior)
Garcia	trailing_4w_std	-1.8	-4.2	133%	⚠ FLAG (demand more volatile recently – something changed Q3)
Kim	attn_recency_bias	0.500	0.413	17%	✓ normal
GARCIA DEEP DIVE (trailing_4w_std drift 133%):					
Q2: trailing_4w_std_case = 0.012 (low volatility – flat history)					
Q3: trailing_4w_std_case = 0.028 (higher volatility – more variable)					
This means Garcia's last 4 weeks were much more variable than usual.					

His demand sometimes hit 150 min/wk, other times dropped to near zero. The Bayesian model has already captured this (σ_{pd} widened from 24.1 to 29.3 min). But the SHAP drift flag adds a layer: the change is recent (last 4 weeks), so it may be a temporary disruption (vacation, case cancellations) rather than a structural shift. Monitor Q4.

POSTERIOR σ_{pd} DRIFT

Provider	Q2 σ_{pd}	Q3 σ_{pd}	drift	Interpretation
Wright	18.9 min	14.1 min	-25%	✓ narrowing as expected (learns)
Garcia	24.1 min	29.3 min	+22%	△ widening (more erratic Q3)
Chen	8.3 min	8.1 min	-2%	✓ stable (no change)

CALIBRATION DRIFT

Level	Q2 coverage	Q3 coverage	drift	Status
50%	53.2%	52.1%	-1.1pp	✓ stable
80%	83.7%	82.4%	-1.3pp	✓ stable
95%	96.1%	95.8%	-0.3pp	✓ stable

Early Warning Cascade

When drift signals fire in sequence, they tell a story:

SHAP drift → **σ_{pd} widening** → **calibration decline** is the full degradation sequence. SHAP drift is the earliest signal (detectable after 1 quarter of unusual data). σ_{pd} widening follows (takes 2 quarters to show in the posterior). Calibration decline comes last (3-4 quarters). Catching it at the SHAP drift stage means corrective action can be taken before the model is seriously miscalibrated.

Corrective actions in order of severity:

1. Note in SUMMARY.md, monitor next quarter
2. Extend historical range to 26 weeks to dilute the anomalous recent data
3. Switch likelihood family for the flagged provider (if the drift signals a regime change, e.g., Normal → ZI-Gamma)
4. Reset Optuna study for the flagged provider and retune XGBoost from scratch

Stage 6: Cross-Layer Narrative Generation

What a Narrative Is

The SHAP codes, coverage curves, counterfactuals, and drift reports are quantitative outputs. A narrative is the single coherent paragraph-length explanation that a committee member would read first, before diving into numbers.

The narrative synthesizes all upstream signals into a plain statement:

- What the engine recommends for this provider
- Why (the dominant SHAP driver)
- Whether the goal was met and how confidently
- The most important risk or opportunity identified

Narrative Template

NARRATIVE · Dr. James Garcia · Q3 2026 · utilization_first

"Dr. Garcia's allocation was reduced from 28 blocks to 6 blocks (from 126 hrs to 48 hrs per quarter). The primary driver is his chronic low utilization history – his attention-weighted operating level of 18% is the single largest negative SHAP contributor (–94.2 min vs a Cardiac peer average of +38.1 min), placing him 132 min below his service line in predicted weekly demand.

With 6 blocks, the engine achieves 87% scenario coverage – above the 85% threshold – making the target utilization goal technically met. However, the utilization band is wide (P10=42%, P50=68%, P90=97%): in a typical quarter Dr. Garcia is likely to reach 68% utilization, but in a quiet quarter he may fall to 42%. This wide band reflects genuine demand uncertainty (posterior σ_{pd} = 29.3 min) amplified by recent volatility – his last 4 weeks showed higher variability than his historical pattern (SHAP drift flag: +133% on trailing_4w_std).

Key opportunity: the What-If analysis shows that if Dr. Garcia reduces his early block release pattern from 316 min/quarter to zero, one fewer block would be needed. This is a behavioral lever worth discussing with Dr. Garcia directly.

```
|| Monitor Q4: if the trailing_4w_std SHAP drift persists into Q4,
|| consider switching Dr. Garcia's likelihood family from Normal to
|| Zero-Inflated Gamma to better capture his intermittent operating
|| pattern."
```

Frontier-Aware Narrative

For each accepted plan, the narrative also explains where the chosen plan sits on the frontier and what was given up:

"The selected plan ($\tau = 0.40$, utilization_first theme) achieves +2.3pp site utilization improvement with 63 block changes. Moving to a more aggressive position ($\tau = 0.70$) would add +0.4pp utilization improvement at the cost of 49 additional block changes. Moving to the stability anchor ($\tau = 0$) would save 29 block changes but recover only 0.7pp less utilization improvement. The knee of the frontier is at $\tau \approx 0.35$, where the trade-off ratio is most favorable."

Interesting Extensions

Narrative consistency check. Run the narrative generation on the previous quarter's plan and compare to the actual outcomes. If the narrative said "Garcia is likely to reach 68% utilization" and the actual was 41%, flag this as a calibration miss and surface it to the model team.

Peer benchmarking narrative. Automatically generate a sentence comparing each provider to their service line peers: "Dr. Garcia is the only Cardiac provider with an underuse_risk_flag. The three other Cardiac providers (Chen, Wright, Rivera) are projected to use their blocks at 91%, 73%, and 78% respectively in the median scenario."

Committee-level synthesis. Generate a site-wide summary narrative: "The proposed plan improves SITE-001 utilization from 28.3% to 30.6% (+2.3pp) by reallocating 149 hours of block time. All three goals are met. The most uncertain allocation is Dr. Garcia (band width 55pp). The greatest opportunity identified this quarter is Dr. Kim — no goal was set but his 14 blocks at 41% utilization suggest a potential future reallocation target."

Layer 3 Artifact Bundle

```
layer3_artifacts/run_YYYYMMDD_HHMMSS/
├─ explanations.json      ← per provider: SHAP + attribution + counterfactuals
│                          + confidence bands + risk profile + narrative
├─ shap_values.pkl        ← raw SHAP matrix (27 features × providers × days)
├─ shap_interactions.pkl  ← SHAP interaction matrix (optional, slower)
├─ counterfactuals.json   ← what-if studio results per goal provider
├─ drift_report.json      ← SHAP drift +  $\sigma_{pd}$  drift + calibration drift
├─ frontier_diagnostics.json ← frontier shape, knee point, quarter-over-quarter
├─ preference_state.json  ← learned  $\epsilon$  values,  $\tau$  history, suggested  $\tau$ 
├─ narratives.json         ← plain-language summary per provider + site
├─ provenance.json        ← links to Layer 2 and Layer 1 run_dirs
└─ SUMMARY.md
```

Next section: Allocation Advisor → Recommendation Layer on Top of Layer 3

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